

PMIC Optimized For Multi-Core High-Performance System

AXP228

Datasheet

Revision 1.0

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X-Powers

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Revision History

Revision	Date	Description
1.0	2013/5/24	Initial version

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Overview

AXP228 is a highly integrated PMIC targeted at single cell Li-battery (Li-ion or Li-polymer) applications that require multi-channel power conversion outputs. It provides an easy and flexible power management solution for multi-core processors to meet the increasingly complex and accurate requirements of power control.

AXP228 comes with an adaptive USB3.0-compatible Flash Charger that supports up to 94% efficiency and 2.2A charge current. It also provides 21 power output channels (including 5-CH DC-DC, with efficiency up to 95%). To ensure the security and stability of the power system, AXP228 provides multiple-channel 12-bit ADC for voltage/current/temperature monitoring and integrates protection circuits such as OVP, UVP, OTP, and OCP. Moreover, AXP228 integrates a unique E-Gauge™ system which simplifies battery power measurement.

In addition, AXP228 contains a fast interface for the system to dynamically adjust output voltage and enable work mode switch in order to optimize battery life.

Importantly, AXP228 also features an IPS™ (Intelligent Power Select) circuit that transparently selects power path among USB, external adaptor, Li-battery, and system load, allowing the system to function normally when only running on external input power and not the battery.

AXP228 is available in 8mm x 8mm x 0.75mm 68-pin QFN package.

Applications

- Tablets, smartphones, smart TVs, DVRs
- UMPC and UMPC-like, student computers

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Features

IPS™

- Input voltage range: 2.9V~6.3V (AMR: -0.3V~11V)
- Configurable IPS™ system
- Adaptive USB/AC adaptor voltage/current limit (4.4V/900mA/500mA)

Flash Charger

- Integrated MOSFET charge current up to 2.2A
- Battery temperature monitor
- Fully supports USB charge
- High charge accuracy, $\pm 0.5\%$ accuracy
- Supports 4.1V/4.2V/4.22V/4.24V battery
- Automatic charge control
- Supports LED to indicate charge status
- Automatic charge current adjustment based on system load

Buck DC-DC Converters (5-CH)

- DC-DC1: 1.6V~3.4V adjustable, 100mV/step, load current up to 1.4A
- DC-DC2: 0.6V~1.54V adjustable, 20mV/step, load current up to 2.5A, supports VRC (Voltage Ramp Control)
- DC-DC3: 0.6V~1.86V adjustable, 20mV/step, load current up to 2.5A
- DC-DC4: 0.6V~1.54V adjustable, 20mV/step, load current up to 0.6A
- DC-DC5: 1.0V~2.55V adjustable, 50mV/step, load current up to 2A

LDOs (14-CH)

- RTC_VCC: 30mA, always valid
- ALDO1/2: low noise LDO, 0.7V~3.3V adjustable, 100mV/step, load current up to 300mA
- ALDO3: low noise LDO, 0.7V~3.3V adjustable, 100mV/step, load current up to 200mA
- LDOIO0/LDOIO1: low noise LDO, 0.7V~3.3V adjustable, 100mV/step, load current up to 100mA
- DLDO1/ ELDO1: 0.7~3.3V adjustable, 100mV/step, load current up to 400mA
- DLDO2/ DLDO3/ ELDO2/ ELDO3: 0.7~3.3V adjustable, 100mV/step, load current up to 200mA
- DLDO4: 0.7V~3.3V adjustable, 100mV/step, load current up to 100mA
- DC5LDO: 0.7V~1.4V adjustable, 100mV/step, load current up to 200mA

Switch (2-CH)

- DC1SW: internal resistance 100mOhm, power sourced from DCDC1
- CHGLED: 100mA sink strength, can be used to drive the motor and charging LED

Host Interface

- P2WI (Push-Pull Two Wire Interface) for host communication
- Configurable interrupt management
- Flexible pin function configuration: 2 GPIOs can be set as IO or LDO, etc
- Integrated timer
- 12 groups of registers for system shutdown data storage

E-Gauge™ System

- Highly accurate gauge system with dual modes
- Easy Mode: highly adaptive to different powers
- Exact Mode: highly accurate data is provided for specific power
- Provides rich power information, such as instantaneous power consumption (mA or mW), remaining power (% or mA), charge status (%), remaining power life, charge time, etc.
- Low power warning and low power protection
- Provides die temperature

System Management

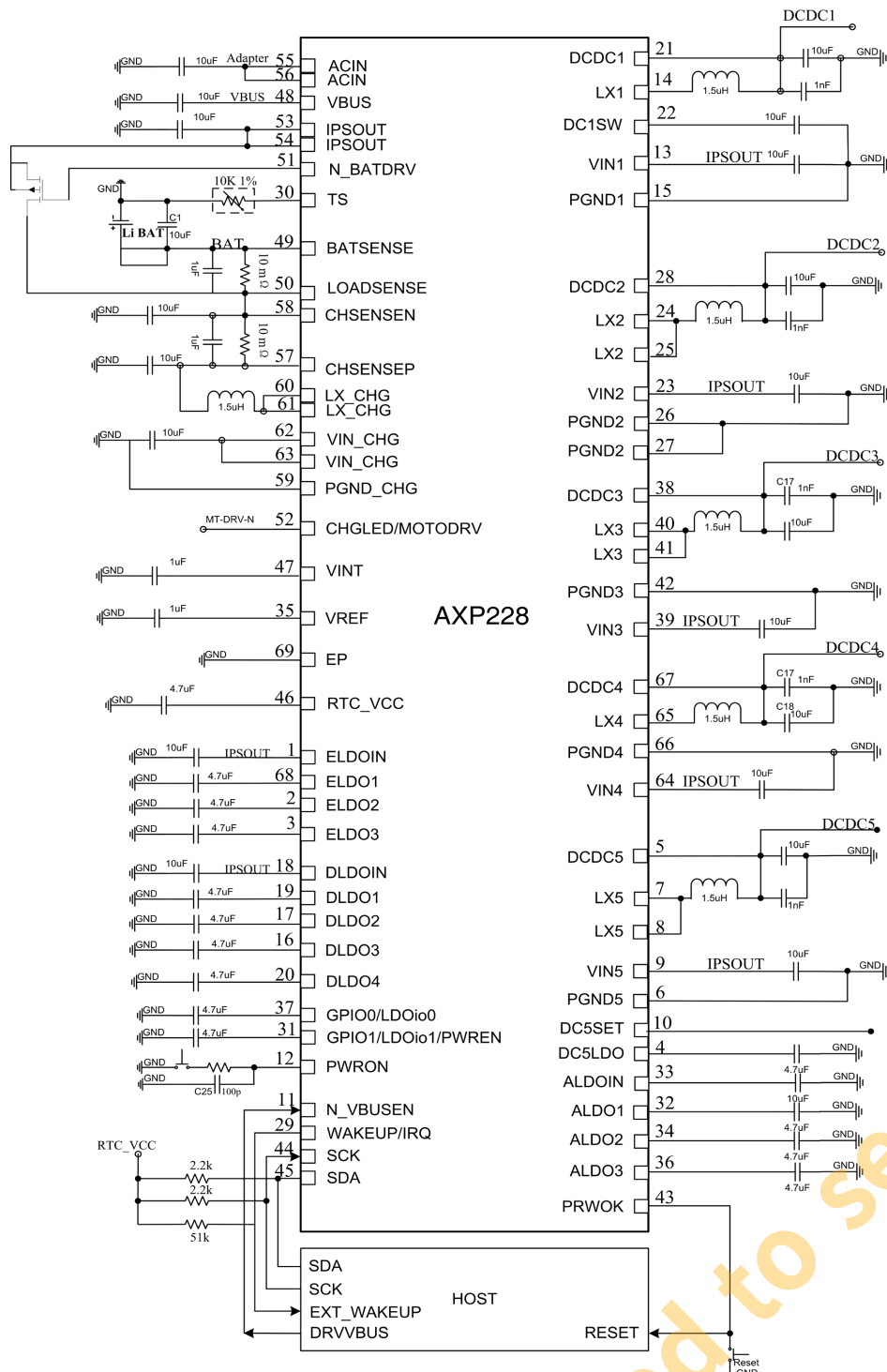
- Supports soft reset and hard reset
- Supports soft shutdown and hard shutdown
- Supports external wakeup triggers
- Supports PWROK for system rest or shutdown indication
- External power detection (insert/remove/drive strength deficiency)
- All output voltage support software boot
- Over/Under-voltage protection (OVP/UVP)
- Over-current protection (OCP)
- Over-temperature protection (OTP)

High Integration

- Highly accurate (0.5%) reference voltage
- Integrates MOSFET

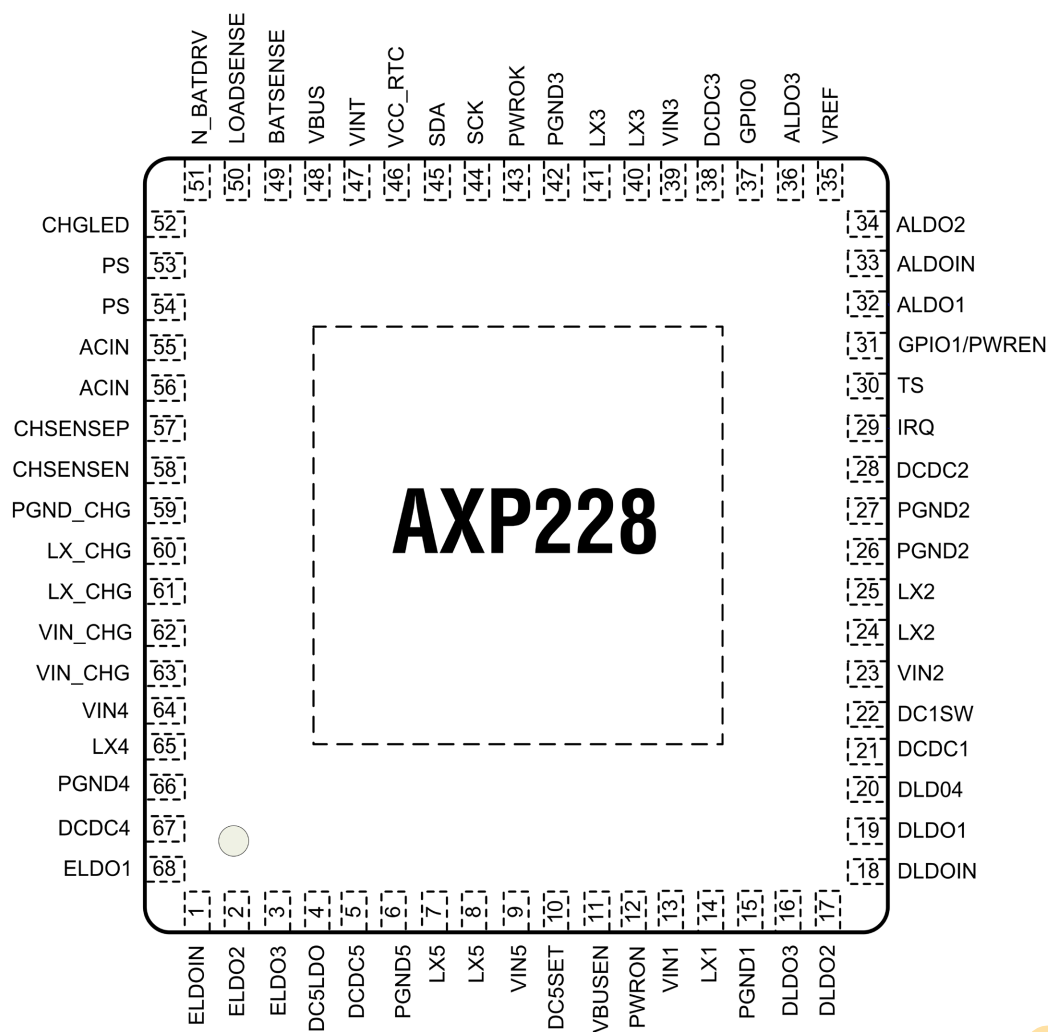
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Typical Applications



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Pin Map



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Pin Description

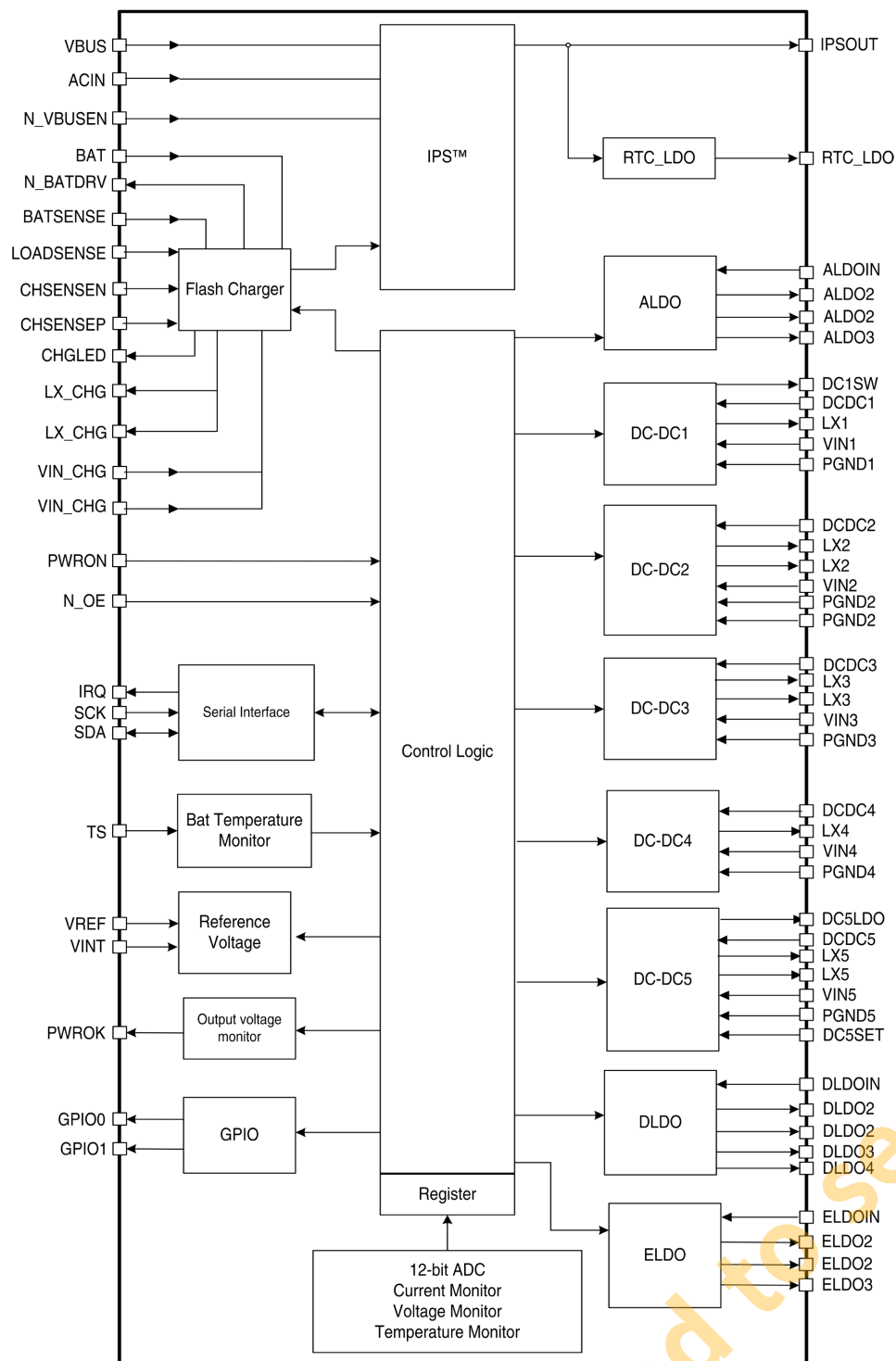
No.	Pin Name	Type	Condition	Description
1	ELDOIN	PI		ELDO Input source
2	ELDO2	O		Output Pin of ELDO2
3	ELDO3	O		Output Pin of ELDO3
4	DC5LDO	O		Output Pin of DC5LDO
5	DCDC5	I		DC-DC5 feedback pin
6	PGND5	G		NMOS Ground for DCDC5
7,8	LX5	IO		Inductor Pin for DCDC5
9	VIN5	PI		DCDC5 input source
10	DC5SET	I		Setting DCDC5 Output Voltage
11	N_VBUSEN	IO	Input	VBUS to IPSOUT Selection GND: IPSOUT selects VBUS High: IPSOUT does not select VBUS
			Output	VBUS to IPSOUT Selection IPSOUT selects VBUS : GND IPSOUT does not select VBUS: High
12	PWRON	I		Power On-Off key input Internal 100k pull high to APS
13	VIN1	PI		DCDC1 Input Source
14	LX1	IO		Inductor Pin for DCDC1
15	PGND1	G		NMOS GND for DCDC1
16	DLDO3	O		Output Pin of DLDO3
17	DLDO2	O		Output Pin of DLDO2
18	DLDOIN	PI		DLDO Input Source
19	DLDO1	O		Output Pin of DLDO1
20	DLDO4	O		Output Pin of DLDO4
21	DCDC1	I		DC-DC1 feedback pin
22	DC1SW	O		DCDC1 Switch Output Pin
23	VIN2	PI		DCDC2 Input Source
24,25	LX2	IO		Inductor Pin for DCDC2
26,27	PGND2	G		NMOS Ground for DCDC2
28	DCDC2	I		DC-DC2 Feedback Pin
29	IRQ/WAKEUP	IO		IRQ Output or Wakeup

No.	Pin Name	Type	Condition	Description
30	TS	I		Battery Temperature Sensor Input or an External ADC Input
31	GPIO1	IO	REG 92H[2:0]	GPIO1 Low noise LDO
32	ALDO1	O		Output Pin of ALDO1
33	DLDOIN	PI		ALDO Input Source
34	ALDO2	O		Output Pin of ALDO2
35	VREF	O		Internal reference voltage
36	ALDO3	O		Output pin of ALDO3
37	GPIO0	IO	REG 90H[2:0]	GPIO0 Low noise LDO
38	DCDC3	I		DC-DC3 feedback pin
39	VIN3	PI		DCDC3 Input Source
40,41	LX3	IO		Inductor Pin for DCDC3
42	PGND2	G		NMOS Ground for DCDC2
43	PWROK	O		Power Good Indication Output
44	SCK	I		Clock pin for serial interface. Normally, it connects a 2.2K resistor to 3.3V I/O power
45	SDA	IO		Data pin for serial interface. Normally, it connects a 2.2K resistor to 3.3V I/O power
46	VCC_RTC	O		Output Pin of VCC_RTC
47	VINT	PO		Internal logic power, 1.8V
48	VBUS	PI		VBUS Input
49	BATSENSE	I		PWM Charger Current Sense Resistance Positive Input
50	LOADSENSE	I		PWM Charger Current Sense Resistance Negative Input
51	N_BATDRV	O		BAT to PS extern PMOS driver
52	CHGLED	O		charger status indication
53,54	IPSOUT	PO		System power source
55,56	ACIN	PI		Adapter input
57	CHSENSEP	I		PWM Charger Current Limite Sense Resistance Positive Input
58	CHSENSEN	I		PWM Charger Current Limite Sense Resistance Negative Input
59	PGND_CHG	G		NMOS Ground for PWM Charger
60,61	LX_CHG	IO		Inductor Pin for PWM Charger
62,63	VIN_CHG	I		Charger Input source
64	VIN4	I		DCDC4 input source
65	LX4	IO		Inductor Pin for DCDC4
66	PGND4	G		NMOS Ground for DCDC4

No.	Pin Name	Type	Condition	Description
67	DCDC4	I		Feed back to DCDC4
68	ELDO1	O		Output Pin of ELDO1
69	EP	G		Exposed Pad, need to be connected to system ground

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Block Diagram



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Absolute Maximum Rating

Parameters	Description	Value	Unit
ACIN	Input Voltage	-0.3 ~ 11	V
VBUS	Input Voltage	-0.3 ~ 11	V
T _J	Operating Temperature Range	-40 ~ 130	°C
T _s	Storage Temperature Range	-40 ~150	°C
T _{LEAD}	Maximum Soldering Temperature (at leads 10sec)	300	°C
V _{ESD}	Maximum ESD stress voltage(Human Body Model)	>4000	V
P _D	Internal Power Dissipation	2700	mW

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Electrical Characteristics

Para.	Description	Condition	Min	Typ	Max	Unit
ACIN						
V_{IN}	ACIN Input Voltage		3.8		6.3	V
I_{OUT}	V_{OUT} Current Available Before Loading BAT	400mV Voltage Drop		3000		mA
V_{UVLO}	ACIN Under Voltage Lockout			3.8		V
V_{OUT}	IPS TM Output Voltage		2.9		5.0	V
R_{ACIN}	Internal Ideal Resistance	PIN to PIN, ACIN to IPSOUT		110		mOhm
VBUS						
V_{IN}	VBUS Input Voltage		3.8		6.3	V
I_{OUT}	V_{OUT} Current Available Before Loading BAT	400mV Voltage Drop		500	900	mA
V_{UVLO}	VBUS Under Voltage Lockout			3.8		V
V_{OUT}	IPS TM Output Voltage		2.9		5.0	V
R_{VBUS}	Internal Ideal Resistance	PIN to PIN, VBUS to IPSOUT		155		mOhm
Battery Charger						
V_{TRGT}	BAT Charge Target Voltage		-0.5%	4.2	+0.5%	V
I_{CHRG}	Charge Current			1200	2200	mA
I_{TRKL}	Trickle Charge Current			10%		I_{CHRG} mA
V_{TRKL}	Trickle Charge Threshold Voltage			3.0		V
V_{RECHG}	Recharge Battery Threshold Voltage	Threshold Voltage Relative to V_{TARGET}		-100		mV
T_{TIMER1}	Charger Safety Timer Termination Time	Trickle Mode		50		Min
T_{TIMER2}	Charger Safety Timer Termination Time	CC Mode		480		Min
I_{END}	End of Charge Indication Current Ratio	CV Mode		10%	15%	I_{CHRG} mA
NTC						
V_{TL}	Cold Temperature Fault Threshold Voltage	Charge	0	2.112	3.264	V
		Discharge		3.226		
V_{TH}	Hot Temperature Fault Threshold Voltage	Charge	0	0.397	3.264	V
		Discharge		0.5		
V_{TE}	NTC Disable Threshold Voltage	Falling Threshold Hysteresis		0.2		V

Para	Description	Condition	Min	Typ	Max	Unit
Off Mode Current						
I_{BATOFF}	OFF Mode Current	BAT=3.8V		35		μA
Logic						
V_{IL}	Logic Low Input Voltage			0.3		V
V_{IH}	Logic High Input Voltage			1.5		V
P2WI						
V_{CC}	Input Supply Voltage			LDO1		V
f_{SCK}	Clock Operating Frequency				2000	KHZ
t_f	Clock Data Fall Time			60		ns
t_r	Clock Data Rise Time			100		ns
DCDC						
f_{OSC}	Oscillator Frequency	Default		3		MHz
DCDC1						
I_{VIN1}	Input Current	PFM Mode $I_{DC1OUT}=0$		50		μA
I_{LIM1}	PMOS Switch Current Limit	PWM Mode		2000		mA
I_{DC1OUT}	Available Output Current	PWM Mode		1400		mA
V_{DC1OUT}	Output Voltage	Default	1.6	3.0	3.4	V
DCDC2						
I_{VIN2}	Input Current	PFM Mode $I_{DC2OUT}=0$		50		μA
I_{LIM2}	PMOS Switch Current Limit	PWM Mode		2800		mA
I_{DC2OUT}	Available Output Current	PWM Mode		2500		mA
V_{DC2OUT}	Output Voltage Range		0.6	1.1	1.54	V
DCDC3						
I_{VIN3}	Input Current	PFM Mode $I_{DC3OUT}=0$		50		μA
I_{LIM3}	PMOS Switch Current Limit	PWM Mode		2800		mA
I_{DC3OUT}	Available Output Current	PWM Mode		2500		mA
V_{DC3OUT}	Output Voltage Range		0.6	1.1	1.86	V
DCDC4						
I_{VIN4}	Input Current	PFM Mode $I_{DC4OUT}=0$		45		μA
I_{LIM4}	PMOS Switch Current Limit	PWM Mode		1200		mA
I_{DC4OUT}	Available Output Current	PWM Mode		600		mA
V_{DC4OUT}	Output Voltage Range		0.6	1.1	1.54	V
DCDC5						
I_{VIN5}	Input Current	PFM Mode $I_{DC5OUT}=0$		45		μA
I_{LIM5}	PMOS Switch Current Limit	PWM Mode		2500		mA
I_{DC5OUT}	Available Output Current	PWM Mode		2000		mA
V_{DC5OUT}	Output Voltage Range		1.0	1.5	2.55	V

Para.	Description	Condition	Min	Typ	Max	Unit
RTC_VCC						
V_{RTC_VCC}	Output Voltage	$I_{RTC_VCC}=1mA$	-1%	3.0	1%	V
I_{RTC_VCC}	Output Current			30		mA
ALDO1						
V_{ALDO1}	Output Voltage	$I_{ALDO1}=1mA$	-1%	OFF	1%	V
I_{ALDO1}	Output Current			300		mA
I_Q	Quiescent Current			60		μA
PSRR	Power Supply Rejection Ratio	$I_{ALDO1}=60mA, 1KHz$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3V, I_O=20mA$		31		μV_{RMS}
ALDO2						
V_{ALDO2}	Output Voltage	$I_{ALDO2}=1mA$	-1%	1.8	1%	V
I_{ALDO2}	Output Current			300		mA
I_Q	Quiescent Current			62		μA
PSRR	Power Supply Rejection Ratio	$I_{ALDO2}=10mA, 1KHz$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3V, I_O=20mA$		31		μV_{RMS}
ALDO3						
V_{ALDO3}	Output Voltage	$I_{ALDO3}=1mA$	-1%	3.0	1%	V
I_{ALDO3}	Output Current			200		mA
I_Q	Quiescent Current			60		μA
PSRR	Power Supply Rejection Ratio	$I_{ALDO3}=10mA, 1KHz$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3V, I_O=20mA$		43		μV_{RMS}
DLDO1						
V_{DLDO1}	Output Voltage	$I_{DLDO1}=1mA$	-1%	OFF	1%	V
I_{DLDO1}	Output Current			400		mA
I_Q	Quiescent Current			56		μA
PSRR	Power Supply Rejection Ratio	$I_{DLDO1}=10mA, 1KHz$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3V, I_O=20mA$		100		μV_{RMS}
DLDO2						
V_{DLDO2}	Output Voltage	$I_{DLDO2}=1mA$	-1%	OFF	1%	V
I_{DLDO2}	Output Current			200		mA
I_Q	Quiescent Current			60		μA

Para.	Description	Condition	Min	Typ	Max	Unit
PSRR	Power Supply Rejection Ratio	$I_{DLD02}=10\text{mA}, 1\text{KHz}$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3\text{V}, I_O=20\text{mA}$		100		μV_{RMS}
DLD03						
V_{DLD03}	Output Voltage	$I_{DLD03}=1\text{mA}$	-1%	OFF	1%	V
I_{DLD03}	Output Current			200		mA
I_Q	Quiescent Current			60		μA
PSRR	Power Supply Rejection Ratio	$I_{DLD03}=10\text{mA}, 1\text{KHz}$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3\text{V}, I_O=20\text{mA}$		100		μV_{RMS}
DLD04						
V_{DLD04}	Output Voltage	$I_{DLD04}=1\text{mA}$	-1%	OFF	1%	V
I_{DLD04}	Output Current			100		mA
I_Q	Quiescent Current			60		μA
PSRR	Power Supply Rejection Ratio	$I_{DLD04}=10\text{mA}, 1\text{KHz}$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3\text{V}, I_O=20\text{mA}$		100		μV_{RMS}
ELD01						
V_{ELD01}	Output Voltage	$I_{ELD01}=1\text{mA}$	-1%	OFF	1%	V
I_{ELD01}	Output Current			400		mA
I_Q	Quiescent Current			55		μA
PSRR	Power Supply Rejection Ratio	$I_{ELD01}=10\text{mA}, 1\text{KHz}$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3\text{V}, I_O=20\text{mA}$		100		μV_{RMS}
ELD02						
V_{ELD02}	Output Voltage	$I_{ELD02}=1\text{mA}$	-1%	OFF	1%	V
I_{ELD02}	Output Current			200		mA
I_Q	Quiescent Current			55		μA
PSRR	Power Supply Rejection Ratio	$I_{ELD02}=10\text{mA}, 1\text{KHz}$		TBD		dB
e_N	Output Noise, 20-80KHz	$V_O=3.3\text{V}, I_O=20\text{mA}$		100		μV_{RMS}
ELD03						
V_{ELD03}	Output Voltage	$I_{ELD03}=1\text{mA}$	-1%	OFF	1%	V
I_{ELD03}	Output Current			200		mA
I_Q	Quiescent Current			55		μA
PSRR	Power Supply Rejection Ratio	$I_{ELD03}=10\text{mA}, 1\text{KHz}$		TBD		dB

Para.	Description	Condition	Min	Typ	Max	Unit
e_N	Output Noise,20-80KHz	$V_0=3.3V, I_0=20mA$		100		μV_{RMS}
DC5LDO						
V_{DC5LDO}	Output Voltage	$I_{DC5LDO}=1mA$	-1%	1.1	1%	V
I_{DC5LDO}	Output Current			200		mA
I_Q	Quiescent Current			40		μA
PSRR	Power Supply Rejection Ratio	$I_{DC5LDO}=10mA, 1KHz$		TBD		dB
e_N	Output Noise,20-80KHz	$V_0=3.3V, I_0=20mA$		100		μV_{RMS}
LDOio0						
V_{LDOio0}	Output Voltage	$I_{LDOio0}=1mA$	-1%	OFF	1%	V
I_{LDOio0}	Output Current			100		mA
I_Q	Quiescent Current			35		μA
PSRR	Power Supply Rejection Ratio	$I_{LDOio0}=10mA, 1KHz$		TBD		dB
e_N	Output Noise,20-80KHz	$V_0=3.3V, I_0=20mA$		40		μV_{RMS}
LDOio1						
V_{LDOio1}	Output Voltage	$I_{LDOio1}=1mA$	-1%	OFF	1%	V
I_{LDOio1}	Output Current			100		mA
I_Q	Quiescent Current			35		μA
PSRR	Power Supply Rejection Ratio	$I_{LDOio1}=10mA, 1KHz$		TBD		dB
e_N	Output Noise,20-80KHz	$V_0=3.3V, I_0=20mA$		40		μV_{RMS}
DC1SW						
R_{DC1SW}	Internal Ideal Resistance	PIN to PIN,DCDC1, DC1SW		160		mOhm
CHGLED						
R_{CHGLED}	Internal Ideal Resistance	$V_{in}=0.3V$		2		Ohm

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Operations and Control

When AXP228 works, the P2WI SCK/SDA pin is pulled up to system IO power, and this interface can be used by Host to access and adjust AXP228 working status.

Note that the external power hereinafter includes ACIN and VBUS input.

9.1. Power on/off and Reset

9.1.1. Power Enable Key

A key can be connected between PWRON and GND to function as a Power Enable Key (PEK) or Sleep/Wakeup key. "Pressing" and "Pressing and Holding" this key will result different reactions from AXP228.

9.1.2. Power On

Power on sources include:

- ACIN and VBUS
- PEK
- IRQ low level

AXP228 can be powered on by pressing and holding PEK for a period of time that longer than "ONLEVEL". In practice, the alarm output signal can be connected to IRQ for AXP228 poweron: when the alarm signal is valid (low level), AXP228 will power on.

9.1.3. Power Off

When PEK is pressed and held for a period of time that longer than "IRQLEVEL", the host will write 1 to REG32H[7] for AXP228 power off. When AXP228 is powered off, all power output except LDO1 will be disabled.

AXP228 will automatically power off in following circumstances:

- low voltage input, auto shutdown for undervoltage protection
- low voltage output due to overload, auto shutdown for overload protection
- High voltage input, auto shutdown for overvoltage protection
- Press and hold PEK for a period of time that longer than OFFLEVEL (6s by default), the system will automatically disable all power output except LDO1

The protection system of AXP228 can protect related components from irreversible damage in case

of application system abnormality.

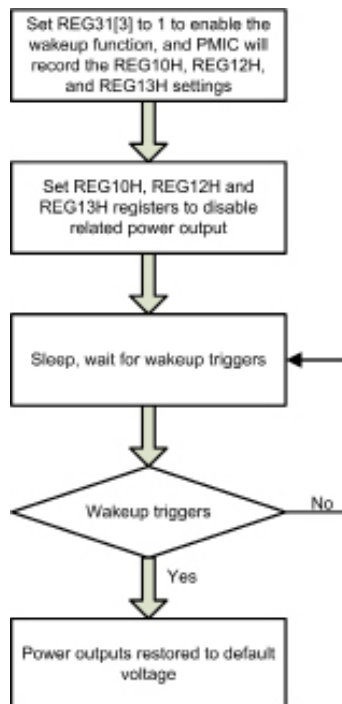
9.1.4. Sleep and Wakeup

To switch from power on mode to sleep mode, several power outputs should be disabled. After that, REG31[3] can be used to control whether following sources can be used to trigger wakeup.

- ACIN connection/disconnection (REG40[6:5] is set to 1)
- VBUS connection/disconnection (REG40[3:2] is set to 1)
- PEK press-and-hold (REG42[0] is set to 1)
- PEK positive edge (REG44[5] is set to 1)
- Battery low power warning Level 2 (REG43[1:0] is set to 1)
- Detection of positive/negative edge when GPIO[1:0] functions input (REG4C[1:0] , REG90[7:6] and REF92[7:6] are set to 1)
- Software wakeup (REG31[5] is set to 1)

After wakeup is triggered, each power output can be restored to default state in right power on sequence.

Here is the Sleep/Wakeup control process:



9.1.5. System Reset and Output Monitoring (PWROK)

The AXP228 PWROK can function as reset signal for application system. The PWROK outputs

low level during the AXP228 power on, and will be pulled high to implement system reset when the output voltage of each channel reaches a preset value. When the application system works, AXP228 will monitor the voltage and load of each channel, and PWROK will output low level to reset the application system if overload or undervoltage occurs to keep from malfunction or data error.

When the application system works, PWROK can also be pulled low to restart AXP228. Each power output will be enabled in certain power on sequence.

9.2. Intelligent Power Select (IPS™)

The AXP228 features Intelligent Power Select IPS™ technology to transparently select power path among Li battery, USB VBUS, and external AC adaptors:

- When only Li-BAT is available, Li-BAT is used for power supply
- If external power (VBUS/ACIN) is available, VBUS/ACIN is used preferentially for power supply
- If external power is disconnected while BATs are available, BATs are used for power supply immediately
- If VBUS and ACIN are connected at the same time, ACIN is used for power supply preferentially, and Li-BAT will be charged; But if the ACIN voltage is too low, VBUS will be used for power supply together with ACIN; if the ACIN+VBUS voltage is too low, the charge current will be reduced to 0A, and Li-BAT will be used for power supply together with them.

Host can visit AXP228 registers through P2WI (Push-Pull Two Wire Interface) to set IPS™.

9.2.1. Voltage-Limited/Current-Limited/Direct Mode

The VBUS is default to work in VBUS Voltage-Limited mode to minimize the influence on USB communication. In Voltage-Limited mode, the VBUS voltage stays above a reference voltage V_{HOLD} , which is 4.4V by default and can be modified in Reg30H[5:3].

If there are limits on the current sourced from USB VBUS, a Current-Limited mode is optional. The current can be set in Reg30H[1:0] as 900mA, 500mA, or unlimited.

VBUS Direct mode can be used by setting REG30H[6] when USB VBUS is used only for power supply or when USB adaptor is used for power supply. When the VBUS is lower than V_{HOLD} , AXP228 will send IRQ to inform that the Host VBUS power current is too weak, in which case the USB communication may be affected, and after that, the Host will react accordingly.

9.2.2. External Power Supply Detection

AXP228 can detect the availability of external power supplies. When external power supplies are connected, AXP228 will automatically judge whether they are usable or not, record the results in corresponding registers, and send out IRQ to inform the Host.

Register Bit	Meaning
REG00_[7]	to indicate whether there is ACIN power supply
REG00_[6]	to indicate whether ACIN is usable
REG00_[5]	to indicate whether there is VBUS power supply
REG00_[4]	to indicate whether VBUS is usable
REG00_[3]	to indicate whether VBUS voltage is above V_{HOLD} when VBUS is connected
REG00_[1]	to indicate whether ACIN/VBUS are short circuited on PCB
REG00_[0]	to indicate whether the system power on is triggered by ACIN/VBUS

When the Host receives IRQ7 (indicating that the VBUS current is too weak), the system will continue to work in Voltage-Limited mode if IRQ7 is caused by system load, or change to Direct mode if IRQ7 occurs because the VBUS voltage is below V_{HOLD} when VBUS is connected.

9.2.3. Select VBUS as Power Supply

N_VBUSEN and REG30H[7],REG30_[2],REG8F_[4] can be set to decide whether use VBUS as the power supply.

REG30_[7]	REG8F_[4]	N_VBUSEN	REG30H[2]	Selected
0	0	High	0	No
0	0	Low	1	Yes
0	1	High	X	No
0	1	Low	X	Yes
1	X	X	X	Yes

Note: when a register bit or status is set to x, it means this bit can be set any value, or the status can be high or low.

9.2.4. Under Current Protection

A automatic shutdown voltage V_{OFF} can be set for system under current protection: if it is lower than ALDOIN, AXP228 will automatically shutdown, and cut disable all power output, except LDO1.

The default of V_{OFF} can be set in REG31H Bit[2:0].

9.2.5. Over Voltage Protection

When the external power voltage exceeds 6.3V, AXP228 will send IRQ1/4 to indicate the over voltage state of external power; when it exceeds 7V, AXP228 will automatically shutdown.

9.3. Adaptive Flash Charger

AXP228 integrates an adaptive Flash Charger to control the charge cycle, adjust charge current, enable BAT detection, trickle charge and activation with alarm and temperature detection circuits.

9.3.1. Adaptive Charge

The charger is ENABLED by default (configurable in REG33H).

When external powers are connected, the AXP228 will automatically check whether they can be used for charge. If there are proper candidates, AXP228 will begin the charge, and send IRQ to the Host to inform it. Meanwhile, CHGLED pin will output low level to drive external LED to indicate the charge status. The diagram right describes the current and voltage during the charge process.

9.3.2. V_{TRET} and T_{RCH}

V_{TRGT} is the target charge voltage, 4.2V by default, and can set in register REG33H[6:5]. It will be adjusted by AXP228 automatically when external power voltage is too low.

V_{RCH} is the voltage threshold when AXP228 will automatically recharge. The V_{RCH} is 0.1V lower than V_{TRGT} .

9.3.3. Charge Current

The charge current is 450mA or 1200mA by default, and it can be set in REG33H[3:0].

9.3.4. Charge Process

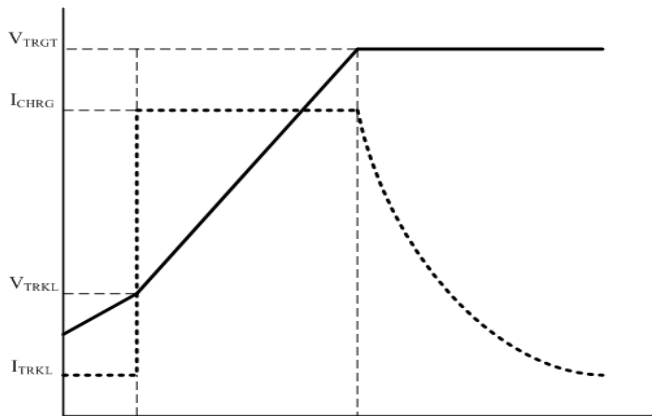
When the battery voltage is lower than 3.0V, the charger will change to pre-charge mode, where the charge current is 1/10 of a preset current value. If the battery voltage fails to reach 3.0V within 40 mins (adjustable in REG34H) the charger will automatically enter battery activation mode.

Once the battery voltage is above 3.0V, the charger will enter constant current charge mode. If the charge current is lower than 65% of the preset current value, IRQ17 will be sent to indicate that the charge capability of external power supply is weak, and a more powerful external power supply

should be used, or power-consuming applications should be killed.

When the battery voltage reaches V_{TRGT} , the charger will go to constant-voltage charge mode, and the charge current will become lower. When the current is reduced to be lower than 10% or 15% (adjustable in REG33H) of the preset current value, the charge cycle finishes. After that, AXP228 will release IRQ13.

A new charge cycle will begin when the battery voltage becomes lower than V_{RCH} , and IRQ12 will be released as well.



9.3.5. Battery Activation Mode

When the charger enters activation mode, either from pre-charge mode or constand-current charge mode (time timeout), AXP228 will release IRQ10.

In activation mode, the battery is charged with low current. When the battery voltage reaches V_{RCH} , the charger will exit from activation mode, and IRQ11 will be released.

REG01H can indicate whether the charger is in battery activation mode.

9.3.6. CHGLED

CHGLED pin can be used to indicate the charge status and for warning as well. The CHGLED is NMOS Open Drain, and it can be used to directly drive a LED through a resistor to indicate the charge status.

Type A :

Status	Reaction	Note
Charge On	low level	
Charge Off	High resistance	
Battery abnormality	25% duty 1Hz flicker	the charger enters battery activation mode, or the battery temperature is too high/low

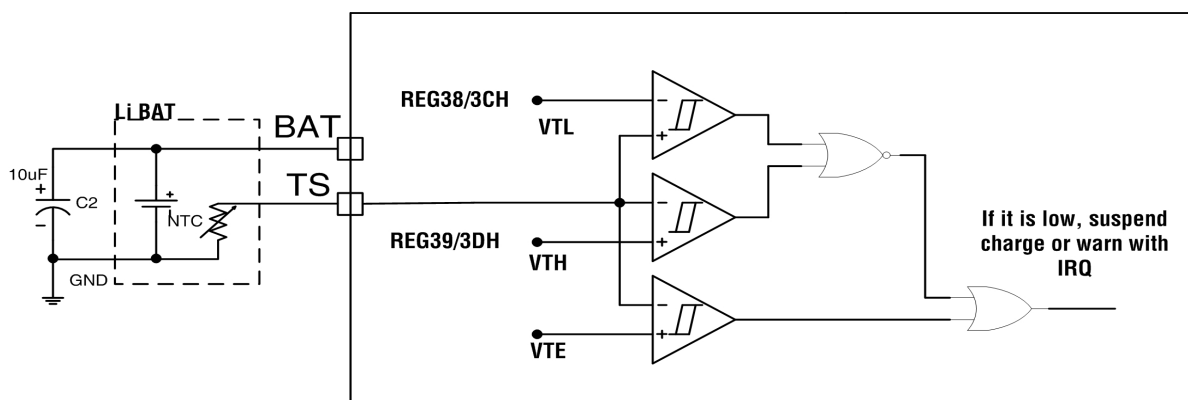
Status	Reaction	Note
Over voltage	25% duty 4Hz flicker	voltage sourced from external power supply is too high

Type B

Status	Reaction	Note
Charge On	25% duty 1Hz flick	
Charge Off	High restistance	
battery abnormality or over voltage	25% duty 1Hz flicker	the charger enters battery activation mode, or the battery temperature is too high/low, or the charger enters battery activation mode, or the battery temperature is too high/low
No battery	low level	no batteries available

9.3.7. Battery Temperature Detection

During the battery charge/discharge, the battery temperature can be detected by AXP228 by connecting a temperature-sensitive resistor to the TS pin. Here is the circuit diagram:



In the diagram above, the VTH/VTL are thresholds for high temperature and low temperature, which can be set in REG38H/39H/3CH/3DH respectively. The VTE is 0.2V. The NTC temperature sensitive resistors of 10KOhm (25 celsius degree) with 1% accuracy are recommended. When the constant current (adjustable in REG84H to 20uA/40uA/60uA/80uA) from TS pin goes through the resistor, a voltage will be generated, which will then be compared with a preset value so that the AXP228 can react accordingly, releasing IRQs or suspending the charge.

In case that resistance of the temperature sensitive resistor is too high or too low, an extra resistor can be serially or parallelly connected to broaden the detect range.

In case that there is no temperature sensitive resistor, the TS pin can be connected to GND, and AXP228 will disable the temperature detect function automatically.

9.3.8. Battery Detection

AXP228 can detect the availability of batteries, and record the result in REG01H, IRQ8 and IRQ9 will be released accordingly.

The battery detection can be enabled/disabled by Host (REG32H).

9.4. Multi-Channel Power Output

AXP228 is capable of multi-channel power output, as shown below:

Output	Type	Default Voltage	Start Step	Application Examples	I _{max}
DCDC1	BUCK	3.3V	1	3.3V I/O	1400mA
DCDC2	BUCK	1.1V	1	1.1V CPU	2500mA
DCDC3	BUCK	1.1V	1	1.1V GPU	2500mA
DCDC4	BUCK	1.1V	1	1.1V Core	600mA
DCDC5	BUCK	1.5/DC5SET	1	1.5V DDR3	2000mA
RTC-LDO	LDO	3.0V	1	RTC	30mA
ALDO1	LDO	OFF	OFF	N/A	300mA
ALDO2	LDO	1.8V	1	N/A	300mA
ALDO3	LDO	3.0V	1	N/A	200mA
LDO _{IO0}	LDO	OFF	OFF	N/A	100mA
LDO _{IO1}	LDO	OFF	OFF	N/A	100mA
DLDO1	LDO	OFF	OFF	N/A	400mA
DLDO2	LDO	OFF	OFF	N/A	200mA
DLDO3	LDO	OFF	OFF	N/A	200mA
DLDO4	LDO	OFF	OFF	N/A	200mA
ELDO1	LDO	OFF	OFF	N/A	400mA
ELDO2	LDO	OFF	OFF	N/A	200mA
ELDO3	LDO	OFF	OFF	N/A	200mA
DC5LDO	LDO	1.1V	1	N/A	200mA
DC1SW	Switch	OFF	OFF	N/A	400mA

AXP228 comes with 5-ch simultaneous step-down DC-DCs, 14-ch LDOs, 2-ch switch as well as

multiple timing and controlling methods. The DC-DC work frequency is 3MHz by default, which is adjustable in registers. Small inductors and capacitors can be connected as well. In addition, all DC-DCs can be set in PWM mode or auto mode (automatically switchable according to the AXP228 load) in REG80H.

9.4.1. DC-DC1/2/3/4/5

DCDC1 outputs 1.6V-3.4V, DCDC2/4 outputs 0.6V-1.54V, DCDC3 outputs 0.6V-1.86V, DCDC5 outputs 1.0V-2.55V, which can be set in registers.

Small ESR ceramic capacitors above 10uF X7R are recommended to be the DCDC output capacitors; 1.5uH inductors are recommended, and the inductor saturation current should be larger than 50% of the largest demanded current in the circuit.

9.4.2. RTC-LDO

LDO1 is always enabled to provide constant power supply for the RTC of applications. Its maximum load current is 30mA.

9.4.3. ALDO1/2/3

ALDO1/2/3 adopts the low noise technology, and are used for analog power supply of application system. Their maximum load currents are 300mA / 300mA / 200mA respectively.

9.4.4. LDO_{IO0} / LDO_{IO1}

LDO_{IO0} / LDO_{IO1} adopts the low noise technology, and their maximum load currents are 100mA.

9.4.5. DLDO1/DLDO2/DLDO3/DLDO4

DLDO1/DLDO2/DLDO3/DLDO4 are general LDOs, and their maximum load currents are 400mA / 200mA / 200mA / 100mA respectively.

9.4.6. ELDO1/ELDO2/ELDO3/

ELDO1/ELDO2/ELDO3 are general LDOs, and their maximum load currents are 400mA / 200mA / 200mA respectively.

9.4.7. DC5LDO

DC5LDO power is sourced from DC-DC5, and its maximum load current is 200mA.

9.4.8. DC1SW

DC1SW power is sourced from DC-DC1 and its equivalent resistance is 160mOhm.

9.4.9. Soft Start

All DC-DCs and LDOs support soft start to avoid the impact of dramatic current change on the input path during system boot stage.

All DC-DCs do not require external Schottky diodes and resistor divider feedback circuits. If a certain DC-DC is unnecessary in application, just float corresponding LX pins.

9.5. E-Gauge™

AXP228 integrates a dual-mode power gauge system: in simple mode, it delivers relatively high compatibility with various batteries; in high precision mode, it is capable of parameter optimization for certain battery and provides up to 2% precision power gauge.

The disabling/enabling and sample speed of each ADC can be set in REG84H, and related results will be stored in registers. Reference can be made to the ADC data registers in next chapter. The battery current direction can be indicated in REG00H[2].

Channel	000H	STEP	FFFH
Battery Voltage	0mV	1.1mV	4.5045V
Bat discharge current	0mA	1mA	4.095A
Bat charge current	0mA	1mA	4.095A
Internal temperature	-267.7 celsius degree	0.1 celsius degree	165.8 celsius degree
TS pin input	0mV	0.8mV	3.276V

9.6. Pin Function Description

GPIO[1:0]:

to function as GPIO[1:0] or LDO, etc. See REG90H-92H for details.

CHGLED

to indicate the charge status, for over-temperature and over-voltage warning, and motor-drive functions. When REG32[2] is 0, the maximum load current of this pin is 100mA and it can drive motors directly when a small motor is connected to 3.3V power with current-limiting resistors connected; when REG32[2] is 1, this pin is used to indicate the charge status and for warning.

9.7. Timer

AXP228 comes with an internal timer, whose value can be modified in REG8AH[6:0]. The lowest unit is minute, and REG8AH[7] will be set after timer timeout.

9.8. Interrupt

Host can visit AXP228 registers through P2WI interface, with speed up to 2MHz, R/W operations supported.

Interrupts can be pulled low by AXP228 to inform the Host of certain events, and the interrupt status will be stored in interrupt registers (REG48H, REG49H, REG4AH, REG4BH, REG4CH), which can also be cleared by writing 1 to corresponding register bits. When no interrupt event occurs, IRQ will be pulled high through external 51K pull up resistor. Every interrupt can be blocked by setting related registers (REG40H, REG41H, REG42H, REG43H, REG44H).

Register	IRQ	Description	Register	IRQ	Description
REG48_[7]	IRQ1	ACIN over-voltage	REG4B_[7]		
REG48_[6]	IRQ2	ACIN is connected	REG4B_[6]		
REG48_[5]	IRQ3	ACIN is disconnected	REG4B_[5]		
REG48_[4]	IRQ4	VBUS over-voltage	REG4B_[4]		
REG48_[3]	IRQ5	VBUS is connected	REG4B_[3]		
REG48_[2]	IRQ6	VBUS is disconnected	REG4B_[2]		
REG48_[1]	IRQ7	VBUS is lower than V_{HOLD}	REG4B_[1]	IRQ20	low power waring 1
REG48_[0]	reserved		REG4B_[0]	IRQ21	low power waring 2
REG49_[7]	IRQ8	battery is connected	REG4C_[7]	IRQ22	timer finishes timing
REG49_[6]	IRQ9	battery is disconnected	REG4C_[6]	IRQ23	PEK rising edge
REG49_[5]	IRQ10	enter battery activation mode	REG4C_[5]	IRQ24	PEK falling edge
REG49_[4]	IRQ11	exit battery activation mode	REG4C_[4]		
REG49_[3]	IRQ12	be in charge status	REG4C_[3]		
REG49_[2]	IRQ13	charge finished	REG4C_[2]	reserved	
REG49_[1]	IRQ14	battery temperature is too high	REG4C_[1]	IRQ25	GPIO1 edge trigger
REG49_[0]	IRQ15	battery temperature is too low	REG4C_[0]	IRQ26	GPIO0 edge trigger

Register	IRQ	Description	Register	IRQ	Description
REG4A_ [7]	IRQ16	internal IC temperature is too high			
REG4A_ [6]	IRQ17	charge current too low			
REG4A_ [5]	reserved				
REG4A_ [4]	reserved				
REG4A_ [3]	reserved				
REG4A_ [2]	reserved				
REG4A_ [1]	IRQ18	PEK press			
REG4A_ [0]	IRQ19	PEK press and hold			

10

Register Description

10.1. Register List

10.1.1. Power Control Register List

Address	Register Description	R/W	Default
00	Power status register	R	
01	Power mode / charge status register	R	
04-0F	Data cache register	R/W	00H
10	DCDC1/2/3/4/5&ALDO1/2&DC5LDO disable/enable register	R/W	BFH
12	ELDO1/2/3&DLDO1/2/3/4&DC1SW disable/enable register	R/W	00H
13	ALDO3 disable/enable register	R/W	01H
15	DLDO1 voltage set register	R/W	00H
16	DLDO2 voltage set register	R/W	00H
17	DLDO3 voltage set register	R/W	00H
18	DLDO4 voltage set register	R/W	00H
19	ELDO1 voltage set register	R/W	00H
1A	ELDO2 voltage set register	R/W	00H
1B	ELDO3 voltage set register	R/W	00H
1C	DC5LDO voltage set register	R/W	00H
21	DC-DC1 voltage set register	R/W	00H
22	DC-DC2 voltage set register	R/W	00H
23	DC-DC3 voltage set register	R/W	00H
24	DC-DC4 voltage set register	R/W	00H
25	DC-DC5 voltage set register	R/W	00H
27	DC-DC2/3 VRC REG	R/W	00H
28	ALDO1 voltage set register	R/W	00H
29	ALDO2 voltage set register	R/W	00H
2A	ALDO3 voltage set register	R/W	00H
30	VBUS-IPSOUT channel set register	R/W	60H
31	Wakeup and V _{OFF} voltage set register	R/W	03H
32	Shutdown, BAT detection, and CHGLED control register	R/W	43H
33	Charge control register 1	R/W	C6H
34	Charge control register 2	R/W	45H
35	Charge control register3	R/W	0EH
36	PEK parameter set register	R/W	5DH
37	DCDC convertor work frequency register	R/W	08H

Address	Register Description	R/W	Default
38	BAT charge low temperature warninregister	R/W	A5H
39	BAT charge high temperature warning register	R/W	1FH
3C	BAT discharge low temperature warning register	R/W	FCH
3D	BAT discharge high temperature warning register	R/W	16H
80	DCDC work mode register	R/W	80H
82	ADC enable register1	R/W	E0H
84	ADC simple rate TS pin control register	R/W	32H
85	TS ADC sample rate register	R/W	00H
8A	Timer control register	R/W	00H
8C/8D	PWREN control register	R/W	00H
8F	Over-temperature shutdown control register	R/W	01H

10.1.2. GPIO Control Register List

Address	Register Description	R/W	Default
90	GPIO0 control register	R/W	07H
91	GPIO0 LDO output voltage set register	R/W	1FH
92	GPIO1 control register	R/W	07H
93	GPIO1 LDO output voltage set register	R/W	1FH
94	GPIO[1:0] signal status register	R/W	00H
97	GPIO[1:0] pull-low control register	R/W	00H

10.1.3. Interrupt Control Register List

Address	Register Description	R/W	Default
40	IRQ enable control register1	R/W	D8H
41	IRQ enable control register2	R/W	FFH
42	IRQ enable control register3	R/W	03H
43	IRQ enable control register4	R/W	03H
44	IRQ enable control register5	R/W	00H
48	IRQ status register1	R/W	00H
49	IRQ status register2	R/W	00H
4A	IRQ status register3	R/W	00H
4B	IRQ status register4	R/W	00H
4C	IRQ status register5	R/W	00H

10.1.4. ADC Data Register List

Address	Register Description	R/W
56	AXP228 temperature monitor ADC data high 8 bits	R

Address	Register Description	R/W
57	AXP228 temperature monitor ADC data low 4 bits	R
58	TS input ADC data high 8 bits monitor BAT temperature by default	R
59	TS input ADC data low 4 bits monitor BAT temperature by default	R
78	BAT voltage high 8 bits	R
79	BAT voltage low 4 bits	R
7A	BAT charge current high 8 bits	R
7B	BAT charge current low 5 bits	R
7C	BAT discharge current high 8 bits	R
7D	BAT discharge current low 5 bits	R

Address	Register Description	R/W	Default
B8	Gauge control register	R/W	C0H
B9	Gauge result	R	64H
E0	Battery capacity set bit[14:8]	R/W	00H
E1	Battery capacity set bit[7:0]	R/W	00H
E6	BAT low power threshold register	R/W	A0H

Note that the battery capacity unit is 1.456mAH.

10.2. Register Description

10.2.1. REG 00H: Input Power Status

Bit	Description	R/W
7	to indicate whether ACIN exists 0:No 1: Yes	R
6	to indicate whether ACIN is usable	R
5	to indicate whether VBUS exists 0: No 1: Yes	R
4	to indicate whether VBUS is usable	R
3	to indicate whether the VBUS voltage is above V_{HOLD} before connected	R
2	to indicate the battery current direction 0: discharge 1: charge	R
1	to indicate whether ACIN and VBUS input are in-short on PCB	R
0	to indicate whether the power on source is ACIN/VBUS or not 0: No 1: Yes	R

10.2.2. REG 01H: Power Work Mode and Charge Status Indication

Bit	Description	R/W
7	to indicate whether AXP228 temperature is too high 0: no 1: yes	R
6	to indicate the charge status 0: not in charging or fully charged 1: be charging	R
5	to indicate whether there is battery connected to AXP228 0: no batteries connected to AXP228 1: the battery is connected to AXP228	R
4	reserved, cannot be changed	R
3	to indicate whether the bat enters activation mode or not 0: not in activation mode 1: in activation mode	R
2-0	reserved, cannot be changed	R

10.2.3. REG 04-0FH: Data Cache

Note that the data will be saved during power on/off as long as there are batteries or external powers.

10.2.4. REG 10H: DCDC1/2/3/4/5&ALDO1/2&DC5LDO Output Control

Bit	Description		R/W	Default
7	ALDO2 enable/disable control	0: disable 1: enable	RW	X
6	ALDO1 enable/disable control		RW	X
5	DC-DC5 enable/disable control		RW	X
4	DC-DC4 enable/disable control		RW	X
3	DC-DC3 enable/disable control		RW	X
2	DC-DC2 enable/disable control		RW	X
1	DC-DC1 enable/disable control		RW	X
0	DC5LDO enable/disable control		RW	X

10.2.5. REG 12H: Power Output Control

Bit	Description		R/W	Default
7	DC1SW enable/disable control	0: disable 1: enable	RW	0
6	DLDO4 enable/disable control		RW	0
5	DLDO3 enable/disable control		RW	0
4	DLDO2 enable/disable control		RW	0
3	DLDO1 enable/disable control		RW	X
2	ELDO3 enable/disable control		RW	0
1	ELDO2 enable/disable control		RW	0
0	ELDO1 enable/disable control		RW	X

10.2.6. REG 13H: power output control

Default: 01H

Bit	Description		R/W	Default
7	ALDO3 enable/disable control	0: disable 1: enable	RW	0
6-0	reserved, cannot be changed		RW	0

10.2.7. REG 15H:DLDO1 Output Voltage Set

Default: 00H

Bit	Description		R/W	Default
7-5	reserved, cannot be changed			
4	DLDO1 Output Voltage Set Bit4	0.7-3.3V,100mV/step	RW	0
3	DLDO1 Output Voltage Set Bit3		RW	0
2	DLDO1 Output Voltage Set Bit2		RW	0
1	DLDO1 Output Voltage Set Bit1		RW	0
0	DLDO1 Output Voltage Set Bit0		RW	0

10.2.8. REG 16H:DLDO2 Output Voltage Set

Default: 00H

Bit	Description		R/W	Default
7-5	reserved, cannot be changed			
4	DLDO2 Output Voltage Set Bit4	0.7-3.3V,100mV/step	RW	0
3	DLDO2 Output Voltage Set Bit3		RW	0
2	DLDO2 Output Voltage Set Bit2		RW	0
1	DLDO2 Output Voltage Set Bit1		RW	0
0	DLDO2 Output Voltage Set Bit0		RW	0

10.2.9. REG 17H:DLDO3 Output Voltage Set

Default: 00H

Bit	Description		R/W	Default
7-5	reserved, cannot be changed			
4	DLDO3 Output Voltage Set Bit4	0.7-3.3V,100mV/step	RW	0
3	DLDO3 Output Voltage Set Bit3		RW	0
2	DLDO3 Output Voltage Set Bit2		RW	0
1	DLDO3 Output Voltage Set Bit1		RW	0
0	DLDO3 Output Voltage Set Bit0		RW	0

10.2.10. REG 18H:DLDO4 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	DLDO4 Output Voltage Set Bit4	RW	0
3	DLDO4 Output Voltage Set Bit3		
2	DLDO4 Output Voltage Set Bit2		
1	DLDO4 Output Voltage Set Bit1		
0	DLDO4 Output Voltage Set Bit0		

0.7-3.3V,100mV/step

10.2.11. REG 19H:ELDO1 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	ELDO1 Output Voltage Set Bit4	RW	0
3	ELDO1 Output Voltage Set Bit3		
2	ELDO1 Output Voltage Set Bit2		
1	ELDO1 Output Voltage Set Bit1		
0	ELDO1 Output Voltage Set Bit0		

0.7-3.3V,100mV/step

10.2.12. REG 1AH:ELDO2 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	ELDO2 Output Voltage Set Bit4	RW	0
3	ELDO2 Output Voltage Set Bit3		
2	ELDO2 Output Voltage Set Bit2		
1	ELDO2 Output Voltage Set Bit1		
0	ELDO2 Output Voltage Set Bit0		

0.7-3.3V,100mV/step

10.2.13. REG 1BH:ELDO3 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		

Bit	Description	R/W	Default
4	ELDO3 Output Voltage Set Bit4	RW	0
3	ELDO3 Output Voltage Set Bit3	RW	0
2	ELDO3 Output Voltage Set Bit2	RW	0
1	ELDO3 Output Voltage Set Bit1	RW	0
0	ELDO3 Output Voltage Set Bit0	RW	0

0.7-3.3V, 100mV/step

10.2.14. REG 1CH:DC5LDO Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-3	reserved, cannot be changed		
2	DC5LDO Output Voltage Set Bit2	RW	1
1	DC5LDO Output Voltage Set Bit1	RW	0
0	DC5LDO Output Voltage Set Bit0	RW	0

0.7-1.4V, 100mV/step

10.2.15. REG 21H:DC-DC1 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	DC-DC1 Output Voltage Set Bit4	RW	1
3	DC-DC1 Output Voltage Set Bit3	RW	0
2	DC-DC2 Output Voltage Set Bit2	RW	0
1	DC-DC2 Output Voltage Set Bit1	RW	0
0	DC-DC2 Output Voltage Set Bit0	RW	1

1.6-3.4V, 100mV/step

10.2.16. REG 22H:DC-DC2 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-6	reserved, cannot be changed		
5	DC-DC2 Output Voltage Set Bit5	RW	0
4	DC-DC2 Output Voltage Set Bit4	RW	1
3	DC-DC2 Output Voltage Set Bit3	RW	1
2	DC-DC2 Output Voltage Set Bit2	RW	0
1	DC-DC2 Output Voltage Set Bit1	RW	0
0	DC-DC2 Output Voltage Set Bit0	RW	1

0.6-1.54V, 20mV/step

10.2.17. REG 23H:DC-DC3 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-6	reserved, cannot be changed		
5	DC-DC3 Output Voltage Set Bit5	RW	0
4	DC-DC3 Output Voltage Set Bit4	RW	1
3	DC-DC3 Output Voltage Set Bit3	RW	1
2	DC-DC3 Output Voltage Set Bit2	RW	0
1	DC-DC3 Output Voltage Set Bit1	RW	0
0	DC-DC3 Output Voltage Set Bit0	RW	1

10.2.18. REG 24H:DC-DC4 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-6	reserved, cannot be changed		
5	DC-DC4 Output Voltage Set Bit5	RW	0
4	DC-DC4 Output Voltage Set Bit4	RW	1
3	DC-DC4 Output Voltage Set Bit3	RW	1
2	DC-DC4 Output Voltage Set Bit2	RW	0
1	DC-DC4 Output Voltage Set Bit1	RW	0
0	DC-DC4 Output Voltage Set Bit0	RW	1

10.2.19. REG 25H:DC-DC5 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	DC-DC5 Output Voltage Set Bit4	RW	0
3	DC-DC5 Output Voltage Set Bit3	RW	1
2	DC-DC5 Output Voltage Set Bit2	RW	0
1	DC-DC5 Output Voltage Set Bit1	RW	1
0	DC-DC5 Output Voltage Set Bit0	RW	0

10.2.20. REG 27H:DC-DC2/3 Dynamic Voltage Adjustment Parameter

Default:00H

Bit	Description	R/W	Default
7-4	reserved, cannot be changed		

Bit	Description		R/W	Default
3	DC-DC3 VRC enable/disable control 0: enable, 1: disable		RW	0
2	DC-DC2 VRC enable/disable control 0: enable, 1: disable		RW	0
1	DC-DC3 VRC control	0: 20mV/15.625us=1.6mV/us 1: 20mV/31.250us=0.8mV/us	RW	0
0	DC-DC2 VRC control	0: 20mV/15.625us=1.6mV/us 1: 20mV/31.250us=0.8mV/us	RW	0

10.2.21. REG 28H:ALDO1 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	ALDO1 Output Voltage Set Bit4	RW	0
3	ALDO1 Output Voltage Set Bit3	RW	0
2	ALDO1 Output Voltage Set Bit2	RW	0
1	ALDO1 Output Voltage Set Bit1	RW	0
0	ALDO1 Output Voltage Set Bit0	RW	0

10.2.22. REG 29H:ALDO2 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	ALDO2 Output Voltage Set Bit4	RW	0
3	ALDO2 Output Voltage Set Bit3	RW	1
2	ALDO2 Output Voltage Set Bit2	RW	0
1	ALDO2 Output Voltage Set Bit1	RW	1
0	ALDO2 Output Voltage Set Bit0	RW	1

10.2.23. REG 2AH:ALDO3 Output Voltage Set

Default:00H

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		

Bit	Description	R/W	Default
4	ALDO3 Output Voltage Set Bit4	RW	1
3	ALDO3 Output Voltage Set Bit3	RW	0
2	ALDO3 Output Voltage Set Bit2	RW	1
1	ALDO3 Output Voltage Set Bit1	RW	1
0	ALDO3 Output Voltage Set Bit0	RW	1

0.7-3.3V, 100mV/step

10.2.24. REG 30H:VBUS-IPSOUT Management

Default:60H

Bit	Description	R/W	Default
7	When VBUS is usable, whether VBUS-IPSOUT channel can be enabled 0:N_VBUSEN pin decides whether to enable this channel 1:VBUS-IPSOUT can be enabled regardless of the N_VBUSEN status	RW	0
6	VBUS V_{HOLD} voltage limit control 0: no voltage limit 1: voltage limit	RW	1
5	V_{HOLD} set bit 2	RW	1
4	V_{HOLD} set bit 1	RW	0
3	V_{HOLD} set bit 0	RW	0
2	DRIVEVBUS output status 0: output low level 1: output high level (IPSOUT)	RW	0
1-0	VBUS current limit 00 -900mA, 01-500mA, 1x- no current limit	RW	X

10.2.25. REG 31H: Wakeup Control and V_{OFF} Voltage

Default:03H

Bit	Description	R/W	Default
7	whether PWROK is pulled low when wakeup 0: not pulled low 1: pulled low	RW	0
6	soft restart: write 1 can restart the PMIC, and then this bit will automatically be cleared	RW	0
5	wakeup: write 1 to this bit will restore every channel output, and then this bit will automatically be cleared	RW	0
4	when wakeup function is enabled, whether IRQ can trigger wakeup and whether it is blocked during the wakeup 0: IRQ can trigger wakeup and is blocked during the wakeup 1: IRQ can work, but cannot trigger wakeup		

Bit	Description	R/W	Default
3	wakeup function enable/disable in sleep mode 0: wakeup disabled 1: wakeup enabled Write 1 to enter sleep mode since this bit will be automatically cleared after set.		
2	V _{OFF} set Bit2	000-2.6V 001-2.7V 010-2.8V	RW 0
1	V _{OFF} set Bit1	011-2.9V 100-3.0V 101-3.1V	RW 1
0	V _{OFF} set Bit0	110-3.2V 111-3.3V	RW 1

10.2.26. REG 32H: Shutdown, Battery Detection and CHGLED Control

Default: 43H

Bit	Description	R/W	Default
7	Write 1 to this bit will disable all AXP228 output, except RTC and the charge module	RW	0
6	disable/enable battery detection 0: disable, 1: enable	RW	1
5-4	CHGLED pin function set	00: high resistance 01: 25% 0.5Hz flicker 10: 25% 2Hz flicker 11: output low level	RW 00
3	CHGLED pin control set	0: controlled by REGREG 32HBit[5:4] 1: controlled by charge function	RW 0
2	every power output disable timing 0: disabled simultaneously 1: contrary to the power on timing	RW	0
1-0	delay of PWROK to the enabling of the last channel power output	00: 8ms 01: 16ms 10: 32ms 11: 64ms	RW 10

10.2.27. REG 33H: Charge Control 1

Default: C6H

Bit	Description	R/W	Default
7	charge function enable/disable control, including internal channel and external channel 0: disable 1: enable	RW	1
6-5	charge target voltage 00: 4.1V 01: 4.22V 10: 4.2V 11: 4.24V	RW	10

Bit	Description	R/W	Default
4	current threshold when a charge cycle ends 0: the charge ends when the charge current is lowed than 10% of the preset value 1: the charge ends when the charge current is lowed than 15% of the preset value	RW	0
3-0	the charge current of internal channel 0000:300mA 0001:450mA 0010:600mA 0011:750mA 0100:900mA 0101:1050mA 0110:1200mA 0111:1350mA 1000:1500mA 1001:1650mA 1010:1800mA 1011:1950mA 1100:2100mA 1101:2250mA 1110:2400mA 1111:2550mA	RW	X

10.2.28. REG 34H: Charge Control 2

Default:45H

Bit	Description	R/W	Default
7	Precharge timeout set bit1	00: 40 min 01: 50min	RW 0
6	Precharge timeout set bit10	10: 60min 11: 70min	RW 1
5	charge output disable/enable after the charge finishes 0:disable 1:enable	RW	000
4	CHGLED type select 0: type A 1:type B		0
3	reserved, cannot be changed		0
2	whether the constant charge voltage changes with the charge current 0: No 1: Yes		0
2	external channel enable/disable during the charge status 0: disable 1: enable	RW	0
1	constant current mode timeout set bit1	00: 6Hours 01: 8Hours	RW 0
0	constant current mode timeout set bit 0	10: 10Hours 11: 12Hours	RW 1

Note that details about the type A/B can be found in the adaptive flash charger part.

10.2.29. REG 35H: Charge Control 3

Default:0EH

Bit	Description	R/W	Default
7-4	reserved, cannot be changed		
3-0	current limit set 0000:300mA 0001:450mA 0010:600mA 0011:750mA 0100:900mA 0101:1050mA 0110:1200mA 0111:1350mA 1000:1500mA 1001:1650mA 1010:1800mA 1011:1950mA 1100:2100mA 1101:2250mA 1110:2400mA 1111:2550mA	RW	1110

10.2.30. REG 36H: PEK Parameters Set

Default:5DH

Bit	Description		R/W	Default
7	power on time set bit1	00: 128mS 01: 1S	RW	0
6	power on time set bit0	10: 2S 11: 3S.	RW	1
5	key hold time set bit1	00: 1S 01: 1.5S	RW	0
4	key hold time set Bit0	10: 2S 11: 2.5S.	RW	1
3	disable/enable of auto power off function when the key press-and-hold time is longer than the power off time 0: disable 1: enable		RW	1
2	disable/enable of auto power on function after auto shutdown as the key press-and-hold time is longer than the power down time 0: disable 1: enable		RW	1
1	power off time set bit1	00: 4S 01: 6S	RW	0
0	power off time set bit0	10: 8S 11: 10S	RW	1

10.2.31. REG 37H:DC-DC Work Frequency Set

Default:08H

Bit	Description		R/W	Default
7	DC-DC and PWM charger frequency spread 0: disable 1: enable			0
6	DC-DC and PWM charger frequency spread 0: 50KHz 1:100KHz			0
5	reserved, cannot be changed			
4	DC-DC 2&3 Poly-phase function enable/disable 0: disable 1: enable			X
3	DC-DC disable/enable frequency bit 3	change by 5% per level, 3MHz by default	RW	1
2	DC-DC enable/disable frequency bit 2		RW	0
1	DC-DC enable/disable frequency bit 1		RW	0
0	DC-DC enable/disable frequency bit 0		RW	0

10.2.32. REG 38H:VLTF-charge Charge Low Temperature Threshold

Default:A5H

Bit	Description		R/W	Default
7-0	the battery low temperature threshold during the charge, M	M*10H, when M is A5H, the threshold voltage will be 2.112V; the voltage can range from 0V to 3.264V	RW	A5H

$$V_{LTF-charge} = M * 10H * 0.0008V$$

10.2.33. REG 39H:VHTF-charge Charge High Temperature Threshold

Default:1FH

Bit	Description		R/W	Default
7-0	the battery high temperature threshold during the charge, N	N*10H, when N is 1FH, the threshold voltage will be 0.397V; the voltage can range from 0V to 3.264V	RW	1FH

$$V_{\text{HTF-charge}} = N * 10H * 0.0008V$$

10.2.34. REG 3CH:VLTF-discharge Discharge Low Temperature Threshold

Default:FCH

Bit	Description		R/W	Default
7-0	the battery low temperature threshold during the discharge, M	M*10H, when M is FCH, the threshold voltage will be 3.226V; the voltage can range from 0V to 3.264V	RW	FCH

$$V_{\text{LTF-discharge}} = M * 10H * 0.0008V$$

10.2.35. REG 3DH:VHTF-discharge Discharge High Temperature Threshold

Default:16H

Bit	Description		R/W	Default
7-0	the battery high temperature threshold during the discharge, N	N*10H, when N is 16H, the threshold voltage will be 0.282V, the voltage can range from 0V to 3.264V	RW	16H

$$V_{\text{LTF-discharge}} = N * 10H * 0.0008V$$

10.2.36. REG 80H:DC-DC Work Mode Select

Default:80H

Bit	Description		R/W	Default
7-5	reserved, cannot be changed			
4	DC-DC5 work mode control	0: switch between PFM/PWM 1: PWM	RW	0
3	DC-DC4 work mode control			
2	DC-DC3 work mode control		RW	0
1	DC-DC2 work mode control		RW	0
0	DC-DC1 work mode control			

10.2.37. REG 82H: ADC Enable

Default:E0H

Bit	Description		R/W	Default
7	BAT voltage ADC enable/disable	0: disable 1: enable	RW	1
6	BAT current ADC enable/disable		RW	0
5	Internal temperature ADC enable/disable		RW	0
4-1	reserved, cannot be changed			
0	TS pin ADC function enable/disable	0: disable 1: enable	RW	1

10.2.38. REG 84H: ADC Sample Rate Set, TS Pin Control

Default:32H

Bit	Description		R/W	Default
7	ADC sample rate set bit 1	10×2 ⁿ sample rates are 100, 200, 400, 800Hz respectively	RW	0
6	ADC sample rate set bit 0		RW	0
5-4	TS pin output current 00:20uA 01:40uA 10:60uA 11:80uA		RW	11
3	reserved, cannot be changed			
2	TS pin function 0: BAT temperature detection 1: external independant ADC input		RW	0
1-0	TS pin current output method	00: disable	RW	1
		01: output current during charge status 10: input during ADC sample 11: enable	RW	0

10.2.39. REG 85H:TS ADC Sample Rate Set

Default: 00H

Bit	Description		R/W	Default
7	TS ADC sample rate set 1	10×2 ⁿ sample rates are 100, 200, 400, 800Hz repectively	RW	0
6	TS ADC sample rate set 0		RW	0
5-0	reserved, cannot be changed			

10.2.40. REG 8AH: Timer Control

Default: 00H

Bit	Description	R/W	Default
7	Timer timeout write 1 will clear this status	RW	0
6-0	set the timeout, measured in minutes write "all 0" will disable this timer	RW	0000000

10.2.41. REG 8CH:PWREN Control 1

Default: 00H

Bit	Description	R/W	Default
7	whether DC-DC1 is controlled by PWREN	RW	0
6	whether DC-DC2 is controlled by PWREN	RW	0
5	whether DC-DC3 is controlled by PWREN	RW	0
4	whether DC-DC4 is controlled by PWREN	RW	0
3	whether DC-DC5 is controlled by PWREN	RW	0
2	whether ALDO1 is controlled by PWREN	RW	0
1	whether ALDO2 is controlled by PWREN	RW	0
0	whether ALDO3 is controlled by PWREN	RW	0

0:Yes 1:No

10.2.42. REG 8DH:PWREN Control 2

Default:00H

Bit	Description	R/W	Default
7	whether DLDO1 is controlled by PWREN	RW	0
6	whether DLDO2 is controlled by PWREN	RW	0
5	whether DLDO3 is controlled by PWREN	RW	0
4	whether DLDO4 is controlled by PWREN	RW	0
3	whether ELDO1 is controlled by PWREN	RW	0
2	whether ELDO2 is controlled by PWREN	RW	0
1	whether ELDO3 is controlled by PWREN	RW	0
0	whether DC5LDO is controlled by PWREN	RW	0

0:Yes 1:No

10.2.43. REG 8FH: Auto Shutdown in Case of AXP228 Overtemperature

Default: 01H

Bit	Description	R/W	Default
7	enable/disable of IRQ pin functions as power on or wakeup trigger 0: disable 1:enable	RW	0
6	ACIN/VBUS In-short set 0: automatically detect 1: controlled by REG8F_[5]	RW	0
5	whether ACIN/VBUS is in-short status 0: not in In-short 1: be in-short	RW	0

Bit	Description	R/W	Default
4	N_VBUSEN pin function 0: output, for DRIVEVBUS (to driver external OTG boost module) 1: input, for N_VBUSEN (to control VBUS)	RW	X
3	reserved, cannot be changed		
2	auto shutdown function IRQ enable/disable in case of AXP228 overtemperature 0: disable 1: enable	RW	0
1-0	reserved, cannot be changed	RW	01

10.2.44. REG 90H:GPIO0 Function Set

Default: 07H

Bit	Description	R/W	Default
7	whether the positive edge triggers IRQ /wakeup when GPIO0 functions input 0: No 1: Yes	RW	0
6	whether the negative edge triggers IRQ /wakeup when GPIO0 functions input 0: No 1: Yes	RW	0
5-3	reserved, cannot be changed		
2	GPIO0 pin function set bit 2	RW	1
1	GPIO0 pin function set bit 1	RW	1
0	GPIO0 pin function set bit 0	RW	1

000: output low
001: output high
010: general input
011: enable low noise LDO
100: disable low noise LDO
101-111: floating

10.2.45. REG 91H: GPIO0 LDO Output Voltage

Default: 1FH

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	GPIO0 LDO output voltage set bit4	RW	0
3	GPIO0 LDO output voltage set bit3	RW	0
2	GPIO0 LDO output voltage set bit2	RW	0
1	GPIO0 LDO output voltage set bit1	RW	0
0	GPIO0 LDO output voltage set bit0	RW	0

0.7-3.3V, 100mV/step

10.2.46. REG 92H:GPIO1 Function

Default: 07H

Bit	Description	R/W	Default
7	whether the positive edge triggers IRQ or wakeup when GPIO0 functions input 0: No 1:Yes	RW	0
6	whether the negative edge triggers IRQ or wakeup when GPIO0 functions input 0: No 1:Yes	RW	0
5-3	reserved, cannot be changed		
2	GPIO0 pin function set bit2	RW	1
1	GPIO0 pin function set bit1	RW	1
0	GPIO0 pin function set bit0	RW	1

000: output low
001: output high
010: general input
011: enable low noise LDO
100: disable low noise LDO
101-111: floating

10.2.47. REG 93H:GPIO1 LDO Output Voltage

Default:1FH

Bit	Description	R/W	Default
7-5	reserved, cannot be changed		
4	GPIO1 LDO Output Voltage Set Bit4	RW	0
3	GPIO1 LDO Output Voltage Set Bit3	RW	0
2	GPIO1 LDO Output Voltage Set Bit2	RW	0
1	GPIO1 LDO Output Voltage Set Bit1	RW	0
0	GPIO1 LDO Output Voltage Set Bit0	RW	0

0.7-3.3V,100mV/step

10.2.48. REG 94H:GPIO[1:0] Input Status Detection

Default:00H

Bit	Description	R/W	Default
7-2			
1	GPIO1 input status	R	
0	GPIO0 input status	R	

0: input low level
1: input high level

10.2.49. REG 97H: GPIO[1:0] Input Pull-Down Enable/Disable

Default:00H

Bit	Description	R/W	Default
7-2	reserved, cannot be changed		

Bit	Description	R/W	Default
1	enable/disable of pull-down resistor when GPIO1 functions input	RW	0
0	enable/disable of pull-down resistor when GPIO1 functions input	RW	0

10.2.50. REG 40H:IRQ enable1

Default:D8H

Bit	Description	R/W	Default
7	ACIN overvoltage IRQ enable/disable	RW	1
6	ACIN connection IRQ enable/disable	RW	1
5	ACIN disconnection IRQ enable/disable	RW	0
4	VBUS overvoltage IRQ enable/disable	RW	1
3	VBUS connection IRQ enable/disable	RW	1
2	VBUS disconnection IRQ enable/disable	RW	0
1	"VBUS is usable but lower than V_{HOLD} " IRQ enable/disable	RW	0
0	reserved, cannot be changed	RW	0

10.2.51. REG 41H:IRQ enable2

Default:FFH

Bit	Description	R/W	Default
7	battery connection IRQ enable/disable	RW	1
6	battery disconnection IRQ enable/disable	RW	1
5	"entering battery activation mode" IRQ enable/disable	RW	1
4	"exiting battery activation mode" IRQ enable/disable	RW	1
3	"be charging" status IRQ enable/disable	RW	1
2	"charge finishes" status IRQ enable/disable	RW	1
1	battery overtemperature IRQ enable/disable	RW	1
0	battery undertemperature IRQ enable/disable	RW	1

10.2.52. REG 42H:IRQ Enable/Disable 3

Default:03H

Bit	Description	R/W	Default
7	AXP228 overtemperature IRQ enable/disable	RW	0
6-2	reserved, cannot be changed		

Bit	Description	R/W	Default
1	key press IRQ enable/disable	RW	1
0	key press-and-hold IRQ enable/disable	RW	1

10.2.53. REG 43H: IRQ Enable/Disable 4

Default:03H

Bit	Description	R/W	Default
7-2	reserved, cannot be changed		
1	"BAT power reaches the warning threshold 1" IRQ enable/disable (warning)	RW	1
0	"BAT power reaches the warning threshold 2" IRQ enable/disable (shutdown)	RW	1

10.2.54. REG 44H: IRQ Enable/Disable 5

Default:00H

Bit	Description	R/W	Default
7	timer timeout IRQ enable/disable	RW	0
6	PEK positive edge triggered IRQ enable/disable	RW	0
5	PEK negative edge triggered IRQ enable/disable	RW	0
4-2	reserved, cannot be changed		
1	GPIO1 input edge triggered IRQ enable/disable	RW	0
0	GPIO0 input edge triggered IRQ enable/disable	RW	0

10.2.55. REG 48H: IRQ Status 1

Bit	Description	R/W	Default
7	ACIN overvoltage IRQ status	RW	0
6	ACIN connection IRQ status	RW	0
5	ACIN disconnection IRQ status	RW	0
4	VBUS overvoltage IRQ status	RW	0
3	VBUS connection IRQ status	RW	0
2	VBUS disconnection IRQ status	RW	0
1	"VBUS is usable but lower than V_{HOLD} " IRQ status	RW	0
0	reserved, cannot be changed	RW	0

10.2.56. REG 49H: IRQ Status 2

Bit	Description	R/W	Default
7	BAT connection IRQ status	RW	0
6	IRQ status for battery disconnection	RW	0
5	IRQ status for "enter battery activation mode"	RW	0
4	IRQ status for "exit battery activation mode"	RW	0
3	IRQ status for "be charging"	RW	0
2	IRQ status for "charge finishes"	RW	0
1	IRQ status for battery overtemperature	RW	0
0	IRQ status for battery undertemperature	RW	0

10.2.57. REG 4AH: IRQ Status 3

Bit	Description	R/W	Default
7	IRQ status for AXP228 overtemperature	RW	0
6-2	reserved, cannot be changed		
1	IRQ status for key press	RW	0
0	IRQ status for key press-and-hold	RW	0

10.2.58. REG 4BH: IRQ Status 4

Bit	Description	R/W	Default
7-2	reserved, cannot be changed		
1	IRQ status for "battery power lower than warning threshold 1"	RW	0
0	IRQ status for "battery power lower than warning threshold 2"	RW	0

10.2.59. REG 4CH: IRQ Status 5

Bit	Description	R/W	Default
7	timer timeout IRQ status	RW	0
6	PEK positive edge trigger IRQ status	RW	0
5	PEK negative edge trigger IRQ status	RW	0
4-2	reserved, cannot be changed		
1	"GPIO1 input edge trigger" IRQ status	RW	0
0	"GPIO0 input edge trigger" IRQ status	RW	0

Note that write 1 to related bit of all IRQ status register will clear the status.

10.2.60. REG B8H: gauge control

Default:C0H

Bit	Description	R/W	Default
7	gauge enable/disable control 0: disable 1: enable	RW	1
6	coulombmeter enable/disable control 0: disable 1: enable	RW	1
5	battery capacity adjustment enable/disable 0: disable 1: enable	RW	0
4	battery capacity adjustment status 0: not adjusted 1: being adjusted	RW	0
3-0	reserved, cannot be changed	RW	0

10.2.61. REG B9H: Battery Power Indication

Default:64H

Bit	Description	R/W	Default
7	whether the battery power is correctly measured 0: No 1: Yes	R	0
6-0	battery power indication	R	64

10.2.62. REG E0H: Battery Capacity Set 1

Default:00H

Bit	Description	R/W	Default
7	whether the battery capacity is configured 0: No 1: Yes	RW	0
6-0	battery capacity set bit[14:8]	RW	64

Battery capacity = Value * 1.456mAh

10.2.63. REG E1H: Battery Capacity Set 2

Default:00H

Bit	Description	R/W	Default
7-0	battery capacity set bit [7:0]	RW	64

10.2.64. REG E6H: Battery Low Power Warning Threshold

Default:A0H

Bit	Description	R/W	Default
7	Battery low power warning threshold 1 0000-1111, 5%-20%	RW	1010

6-0	Battery low power warning threshold 2 0000-1111, 0%-15%	RW	0000
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Package

